Building GPS orbits from station data

What we set out to do, what we built, what worked, what did not, and what comes next

customKF project narrative | Revision 5 | 8 October 2026 UTC

What this is. A retrospective proof of concept: our own Python GPS orbit estimator, run on one Windows PC.

- **Inputs:** about 25 public IGS stations, the broadcast navigation message, published models and calibrations.
- **Output:** 24-hour orbits for the eligible healthy GPS satellites with adequate navigation support.
- **The bar:** be closer to IGS final than the NGA precise ephemeris (NGA PE) is. IGS final is a reference, not truth.

It is not a demonstration of general superiority to NGA, of absolute accuracy, or of rapid operation (page 22).

Five fresh dates (Jun 6, 20, 23, Jul 6, 23)	Mean URE	Mean ratio to NGA	Dates below NGA
Our default configuration	3.06 cm	0.766	5 of 5
+ TU Wien gradients (final latency only)	3.00 cm	0.752	5 of 5
+ gradients, 50 stations instead of 25	2.55 cm	0.638	5 of 5
+ gradients, 17 stations near NGA's sites	4.83 cm	1.215	1 of 5
NGA PE, same satellite-epochs	4.01 cm	1	-

- **Default:** below NGA in URE and 3D on all five dates (repaired-baseline validation, page 12), and at 92.0% of 15-minute epochs over six fresh dates including Jun 27 (92.7% with gradients).
- **Station tests:** doubling the network cut URE 15% against 25 stations; 17 stations no longer beat NGA (page 20).

> These are retrospective results on exposed summer dates. Every input was downloaded after the fact and no rapid-deadline replay has been done. The gradient and station-test arms are final latency.

Since Revision 3 the 3D RMS fell from 0.544 m (6-hour arc) to 8.97 cm (24 hours; five-date mean, URE 3.06 cm). Dates, arcs and settings differ, so that is progress, not one controlled contrast.

Reading map: goal and system 2-6; history 7-11; validation and NGA 12-16; rapid suitability 17-18; station tests 19-20; engineering 21; scope 22; limits and next steps 23; evidence 24-25.

1. The goal, the bar and the metrics

The product

GPS-only orbits estimated from station observations (OBS) and broadcast navigation (NAV), with inputs that could plausibly be available within about a day (a "rapid" product). Our own estimator does the work; no GNSS processing package is used. It does rely on external coordinates, calibrations, Earth orientation and model data (page 22). IGS describes its rapid orbits as available about 17 hours after the day ends. R1, IGS products.

The bar: NGA PE against IGS final

NGA's precise ephemeris is an independent operational product. Our target is to be closer to IGS final than NGA PE is, on exactly the same satellite-epochs. IGS final is a combined, high-quality reference, but it is not truth; "closer to IGS final" means closer to that reference.

How the modern comparisons are scored

This applies to the 24-hour comparison series from 20 September onward. Earlier history (pages 7-8) used shorter windows and sometimes IGS rapid, and is labelled where it appears.

- **Comparison.** Raw positions at IGS final's native 15-minute epochs, 00:00:00-23:45:00 GPST. No alignment, no mean removal, no satellite dropped for scoring badly.
- **Population.** Only satellite-epochs present in every arm being compared and in NGA PE.
- **Frame.** Each solution is rotated to Earth-fixed axes with the Earth-orientation model it was fitted with.
- **3D RMS** is the RMS of the full position error.
- **Orbit-only URE** is sqrt(mean(0.98^2 R^2 + (A^2 + C^2)/49)), with R, A, C the radial, along-track and cross-track errors in the reference orbit's axes. Clocks are not included: ours are nuisance parameters.
- **Ratio to NGA** = our URE / NGA's URE on the same keys. Below 1 means closer to IGS final than NGA.
- **Epoch share:** the fraction of 15-minute epochs at which our constellation URE (the RMS over all satellites at that epoch) is below NGA's.
- **Satellite-epoch share:** each satellite at each epoch counted separately. The two shares aggregate differently and need not order the same way.

Production scores come from `Opus_Fork/evaluate.py`. A separately written scorer, `validation/oracle/`, reproduces them within 0.1 mm (page 21).

2. How the system works

The pipeline

1. **Fetch.** Daily 30-second RINEX 3 observation files for each station, and the broadcast navigation for the day and the previous day, from the IGS data centre at BKG. Static models and calibrations are fetched once.
2. **Prepare.**
   - Choose a dual-frequency signal pair; track phase arcs on the native 30 s data; form ionosphere-free code and phase every 600 s above 10 degrees elevation.
   - Apply station coordinates, tides, ocean loading and antenna calibrations.
   - Select healthy broadcast records; compute each satellite's starting state from the broadcast message.
   - Freeze everything, hashed, into one fixture.
3. **Fit.** A batch least-squares estimator, iterated to convergence, with each iteration as a separate guarded worker:
   - integrate every satellite's orbit and its sensitivities over 24 hours;
   - model every observation;
   - fold the observations, chunk by chunk, into a square-root information system;
   - solve and line-search.
4. **Score** against IGS final, with NGA PE beside it.
5. **Export** an SP3 file through a verified temporary copy; an existing product is never overwritten.

Why a batch square-root estimator

A batch fit uses all the data over the arc at once. It exposes weak directions, prior dependence and parameter coupling, which matter with a small network. Square-root (QR) processing avoids squaring the condition number, as normal equations would. Receiver and satellite clocks are eliminated epoch by epoch, so the retained system stays small. For large networks an option folds each phase ambiguity out once its arc has ended (page 19).

A Kalman filter or square-root information filter can represent the same information sequentially. We kept the batch form for its diagnostics; this choice was not tested against a filter. R2, GROOPS cookbook: GNSS network processing.

What runs where

- **Code.** `customkf/` holds the core physics, time and frames. `claude/roadmap/` holds the 24-hour streaming estimator. `Opus_Fork/` holds the production pipeline (fetch, store, preparation, fit controller, models, evaluator, exporter). `validation/oracle/` holds the independent scorer.
- **How jobs run.** Numerical campaign work runs under resource guards on one Windows PC, as resumable chains, because the PC is switched off overnight (page 21).
- **Further reading.** Setup: `SETUP.md`. The technical specification: `docs/REPRODUCING_WITH_AN_LLM.md`.

3. Models used: forces

Force	Model in the current default
Gravity	EGM2008 to degree and order 12; solid-Earth tides from IERS 2010 (anelastic Love numbers, degrees 2-3) [R3, R9]
Sun and Moon	JPL DE440s positions, direct minus indirect attraction [R8]
Relativity	Schwarzschild term
Solar radiation	ECOM5: a nominal Sun-direction acceleration plus five estimated terms per satellite per arc (page 6). Earth shadow as a penumbra cone [R4]
Earth radiation	Uniform Earth: 238 W/m2 infrared and 0.3 albedo with a Lambert phase law. Absorption only, on published box-wing areas
Antenna thrust	Transmit power / (mass x c), radially outward, from the IGS satellite metadata
GPS-IIIA mass	2,600 kg, the published effective value [R7] (metadata: 2,161 kg)
Attitude	Nominal yaw steering; no eclipse yaw manoeuvres

Integration. DOP853 (SciPy) with the state transition and force sensitivities integrated alongside: maximum step 60 s, relative tolerance 1e-12. Synthetic two-body tests agree to 0.5 micrometres over 24 hours, and tested derivatives agree with finite differences to about 13 micrometres. These bound the tested cases only; the full adopted model has not been reintegration-checked at its real boundaries (page 23).

Earth orientation. IERS daily EOP, plus the IERS 2010 sub-daily ocean-tide model (ORTHO_EOP) for polar motion and UT1 [R3]. It is applied in the orbit dynamics, the station geometry, the evaluator and the exporter alike. Libration terms are omitted.

What is fixed and what is estimated

Fixed:

- the Earth-radiation and thrust forces, computed from published geometry and metadata, never fitted;
- the nominal solar-pressure acceleration.

Estimated per satellite:

- the six initial position and velocity corrections to the broadcast-derived start;
- the five solar-pressure terms.

The estimated terms have no priors. The position and velocity priors (1 km, 1 m/s) are deliberately loose.

Omitted:

- frequency-dependent and ocean tides in the gravity field;
- eclipse attitude;
- reflection in Earth radiation;
- empirical along-track or cross-track accelerations.

These omissions are candidates for the remaining along-track error, not identified causes.

4. Observations and estimation

Element	Model
Observables	Ionosphere-free code (C1W/C2W) and carrier phase (L1/L2), every 600 s, 10-degree cutoff
Geometry	Light time iterated in the inertial frame; Shapiro delay; relativistic clock correction
Antennas	IGS20 ANTEX phase-centre offsets and variations for satellites and receivers; phase wind-up
Troposphere	Saastamoinen hydrostatic delay from standard pressure; Niell mapping; wet delay estimated per station every 3 h
Station motion	IGS20 coordinates and velocities; solid-Earth tide; pole tide; FES2014b ocean loading [R10]
Weights	Code 0.9 m / sin(e). Phase 9 mm / sin(e)^1.3 ("elw13", adopted 27 Sep)
Clocks	Receiver and satellite clocks per epoch, eliminated before the solve; BRUX is the clock datum
Ambiguities	Float, one per phase arc; no integer resolution
New phase arc	New satellite; a gap over 60 s in the native data; loss of lock or half-cycle flags; receiver event or slip records; geometry-free or Melbourne-Wubbena jumps

Iterating to a solution

Starting point. Each fit starts from the broadcast orbit at midnight. The broadcast position refers to the antenna phase centre, so it is converted to the centre of mass.

Each iteration:

1. Integrate.
2. Linearize the observations.
3. Accumulate them in chunks of 12 epochs, each a separate guarded worker.
4. Solve for a Gauss-Newton step.
5. Try step fractions 1, 1/2, 1/4 and 1/8 with full reintegration, and keep the first that lowers the cost.

The fit stops when the step moves the orbit by less than 1 mm and the predicted cost reduction is negligible. Typical fits converge in three or four iterations, about an hour of guarded computing for 25 stations. That is a local stationarity rule; it certifies neither the physics nor a unique optimum.

The current default, in one line

`erp+thrust+iiia2600+sdeop+elw13` with ECOM5 (`unit-ecom5.json`): 25 stations, 24-hour arc, previous-day navigation, satellites kept whenever navigation allows. Every element was adopted under its own written rule; page 10 shows how much each one helped.

Optional model (adopted 3 Oct, final latency only): add TU Wien GRAD operational tropospheric gradients (page 13).

5. Once- and twice-per-rev terms

The solar-radiation model is ECOM (the Empirical CODE Orbit Model) [R4]. In our code **D** is the unit vector from the satellite to the Sun, **Y** = r x D normalised (r the unit position vector), and **B** = D x Y. The angle **u** is measured in the orbit plane from the ascending node in ECOM5, and from the Sun's projection on the orbit plane in the seven-term variant. These are separate from the antenna attitude conventions.

Term	Acceleration	Period	Status
D0, Y0, B0	Constant along D, Y, B	-	Estimated in every fit
Bc1, Bs1	cos u and sin u, along B	**once per revolution**	**Estimated in every fit since the 24 h arcs, for every satellite, with no prior**
Dc2, Ds2 (ECOM-7)	cos 2u and sin 2u, along D	**twice per revolution**	Tested twice, rejected both times
Along-/cross-track 1/rev	Empirical once-per-revolution pulses	once per revolution	Never used

Are they added every time?

- **Yes, for the five ECOM5 terms**, including the two once-per-revolution terms. They are estimated anew for every satellite on every 24-hour arc.
- **Earlier arcs used less.** The 6-hour fits of the first phase estimated only D0, or nothing. On those arcs the once-per-revolution terms are poorly separable from the starting state.
- **No twice-per-revolution term, and no along-track or cross-track empirical acceleration, is in any adopted configuration.**

What the twice-per-revolution tests showed

Our orbits had a 12-hour radial "swing" against IGS. A twice-per-revolution D-term was one candidate; a 12-hour pattern does not identify its cause uniquely.

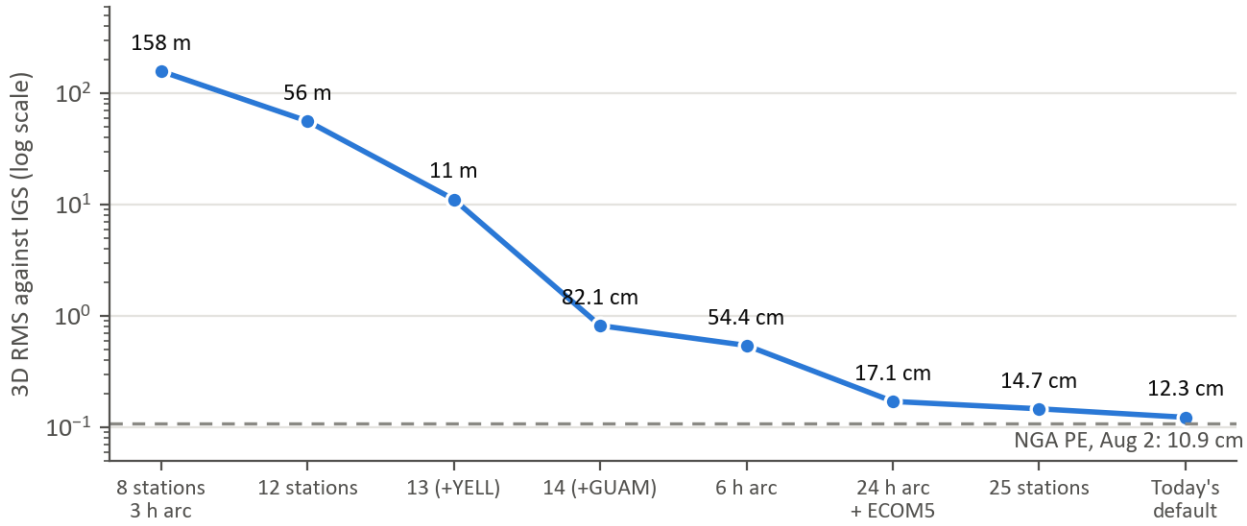
- **Free ECOM-7:** the swing fell 30-37%, but URE worsened 29.5% (Aug 2) and 16.9% (Aug 3). Rejected.
- **ECOM-7 with 2 nm/s2 priors:** Aug 2-5 improved 2-6%, but Aug 7 worsened 4.5% and the hold-out rule failed. The prior width came from fits to IGS orbits, so it was also reference-informed. Rejected.
- **Adding the IERS sub-daily Earth rotation** cut URE by about 14% across the studied days and reduced the 12-hour amplitude on Aug 2-5. On Aug 7 the amplitude rose from 0.47 to 1.07 cm. It was a major contributor on those dates, not a proven sole cause (page 10).

What remains: a residual 12-hour scale "breathing" of 0.16-0.37 ppb, which NGA PE also shows against IGS. A box-wing solar model was implemented but never validated in the station fit.

Evidence: `ECOM7_RESULTS.md`, `ECOM7P_RESULTS.md`, `ECOM7P_HOLDOUT_RESULTS.md`, `SDEOP_RESULTS.md`, in `Opus_Fork/results/`.

6. First phase: 158 m to 15 cm

6-13 September 2026.



Milestones of the first phase on 2 August 2026, 3D RMS against IGS on a log scale. The points are not one matched series: arc length, comparison grid and reference (IGS final or rapid) changed along the way. They show the scale of the progress, not exact step effects.

Step	3D RMS	What it taught us
First station-fitted orbits, 8 stations, 3 h	158 m	A 7 mm phase residual came with a 158 m orbit. Weakly viewed satellites absorb errors
Starting-state fix (antenna phase centre to centre of mass)	157 m	A real bug: start error 27 m to 1.2 m, but not the cause of the poor fit
12, then 13 (+YELL), then 14 stations (+GUAM)	56 m, 11 m, 0.82 m	**Geometry was the largest single gain**: which satellites are seen together, from where
6-hour arc instead of 3	0.54 m	Longer tracking separates orbit from nuisance parameters
24-hour arc + ECOM5	0.17 m	Long arcs let all five solar-pressure terms be estimated
25 stations (greedy geometry screen)	0.147 m (URE 4.6 cm)	The network every later result uses

The 25-station network was chosen by a greedy information screen that added 11 stations to the original 14:

- the original 14: BRUX, HOB2, KOUR, ABMF, KOUG, DJIG, NNOR, MAS1, DAV1, FALK, FAA1, GLPS, YELL, GUAM;
- added: SGOC, KOKB, SUTH, KIT3, KIRI, NKLG, MDO1, EUR2, VACS, PIMO, CHPI.

At that point we were at NGA's level in URE (4.63 against 4.43 cm on Aug 2), but well behind it in 3D (14.7 against 10.8 cm).

Evidence: `STAGE5_RESULTS.md`, `STAGE5B_RESULTS.md`, `YELL_GUAM_RESULTS.md`, `ARC_LENGTH_RESULTS.md`, `claude/roadmap/STEP3_RESULTS.md`, `claude/roadmap/STATIONS_RESULTS.md`, `Opus_Fork/results/BASELINE.md`.

7. Early lessons and failures

Things that did not work, or did not replicate

Experiment (6-hour arcs, Aug 2-3)	Outcome
Estimating one solar-pressure constant (D0) per satellite	Aug 2: 0.544 to 0.442 m (-19%). **Aug 3: 0.621 to 0.713 m (+15%).** Did not replicate; one satellite (G27) dominated
Removing the late GUAM gap on Aug 2 (control)	Did not reproduce the Aug 3 failure; G27 was not explained by that gap
Hourly wet-delay nodes	0.3% gain with six large regressions; not adopted
A 15th station (PIMO) at 3 h	0.3% gain; later part of the 25
Uniformly stronger phase weights	0.962 m against 0.821 m; worse
Block-mean D0 calibrated from IGS rapid	Large gains on 6 h arcs, but **reference-informed**; superseded by 24 h + ECOM5

What we learned

- **Small residuals do not mean a good orbit.** With weak geometry, clocks, ambiguities and weak orbit directions absorb errors.
- **One good day is not a result.** The D0 failure on Aug 3 is why every later change needed a second day, and finally untouched dates.
- **Checks bound errors; they do not identify causes.** Independent checks bounded the tested numerical errors. Fitting ECOM5 directly to IGS positions over 6 hours left 0.8 cm, so the force basis could approximate those arcs closely, but that did not identify the causes of the station-fit error.
- **Later defects still happened.** An antenna-derivative defect, found on 1 October, needed repair and separate validation (page 12).

The comparison rule that came out of it

- Change one thing at a time against a matched same-day arm.
- Write the prediction and the pass/fail rule before the fit.
- Score all satellites, with no exclusions.

From 20 September the rule was enforced: protocols are pinned into each campaign's records before it runs.

Evidence: `D0_POD_RESULTS.md`, `D0_DAY215_RESULTS.md`, `GUAM_GAP_RESULTS.md`, `WET_DELAY_EXPERIMENT.md`, `FORCE_PARAMETER_RESULTS.md`, `claude/roadmap/BLOCK_D0_RESULTS.md`.

8. Second phase: past NGA

From 20 to 27 September a separate code line (`Opus_Fork/`) ran 24-hour fits against IGS final with NGA PE on the same keys. Each step below was one matched same-day change. The URE values are in cm.

Step (adopted)	Aug 2	Aug 3	Aug 4	Aug 5	Aug 7
Baseline	4.427	4.263	4.331	4.225	3.557
+ Earth radiation, antenna thrust (and IIIA mass)	4.167	3.876	4.045	3.836	3.403
+ IERS 2010 sub-daily EOP	3.401	3.165	3.364	3.249	3.409
+ elevation weighting elw13 = default	3.299	3.035	3.218	3.199	3.272
NGA PE	4.427	4.289	4.359	4.826	4.195

A lineage, not one fixed configuration. The IIIA mass was 2,161 kg on Aug 2 and 7 and 2,600 kg on Aug 3-5. Aug 4 and 5 had 23 stations (no CHPI, no YELL); the other days 25. Aug 7 used a different preparation (26 satellites).

Steps before the baseline (Aug 2): the first 25-station 24 h fit scored 4.63 cm URE with 27 satellites; previous-day and per-epoch navigation eligibility recovered 31 (G13 is always excluded as unhealthy); that 31-satellite baseline tied NGA (4.427 against 4.427).

What each step was:

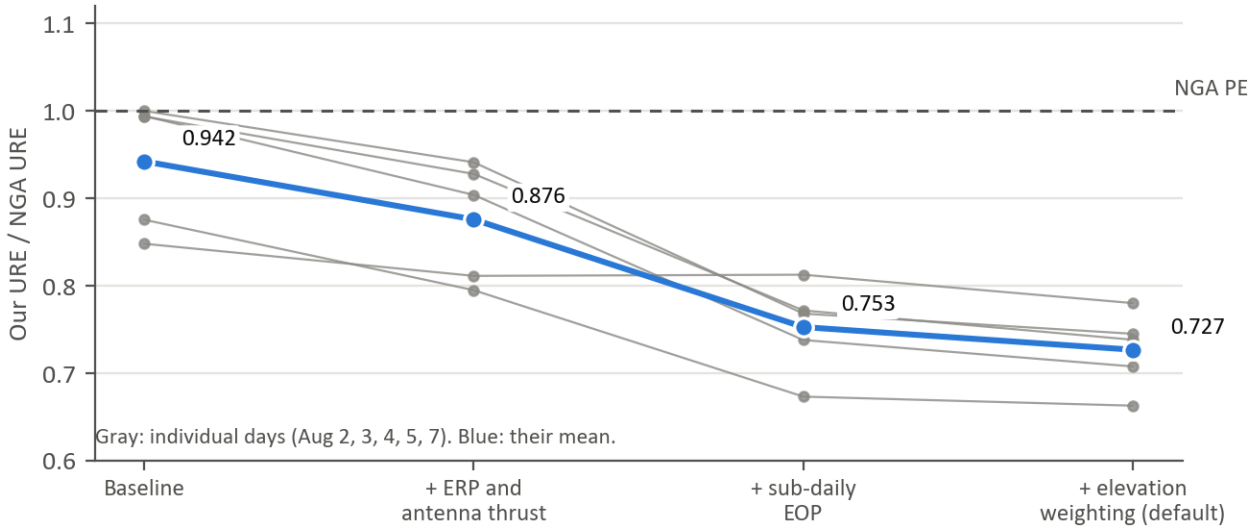
- **Earth radiation and antenna thrust:** fixed physical forces from published geometry and metadata, with nothing fitted. Aug 3-5 also used the published effective IIIA mass, so that step is a composite.
- **Sub-daily EOP:** the IERS 2010 ocean-tide model for polar motion and UT1. It was missing at first, and adding it was the largest single gain (page 6).
- **elw13:** phase sigma 9 mm/sin(e)^1.3. The exponent came from our own post-fit residuals on Aug 3 (1.27); it was not taken from IGS. Choosing to adopt it used IGS-scored outcomes, a selection effect we disclose.

After the last step, URE is below NGA's on all five days, and 3D on four of five. Aug 2 3D is the exception, because of its late-day along-track error.

The limit of these days: all five had already been studied, so these are "new-arm contrasts on previously studied days". A transfer claim needed untouched dates (page 12).

Evidence: `Opus_Fork/results/` (`T2_1_ERP_RESULTS.md`, `IIIA_AUG345_RESULTS.md`, `SDEOP_RESULTS.md`, `ELEVATION_RESULTS.md`), and `docs/consolidation/configurations.md`.

9. How much each improvement helped



Our URE divided by NGA's URE through the adopted steps, for the five studied August days (gray) and their mean (blue). Each step is a matched same-day change; below the dashed line we are closer to IGS final than NGA PE is.

Improvement	URE change	Days	Status
25 stations instead of 14 (24 h, ECOM5)	3D 17.1 to 14.6 cm (vs IGS rapid)	Aug 2	adopted
Previous-day + per-epoch navigation (27 to 31 satellites)	full constellation	Aug 2	adopted
Earth radiation + antenna thrust	-5.9% (Aug 2), -4.3% (Aug 7)	2	adopted
+ published IIIA mass 2,600 kg (composite with the above)	-8.3% mean	Aug 3-5	adopted
IERS sub-daily EOP	-13.7% mean (-18.4 to +0.2%)	5	adopted
Elevation weighting elw13	-3.2% mean, held-out days	4	adopted
TU Wien GRAD gradients (optional)	-4.2% Aug 3; -1.9% mean on 5 fresh dates; -3.8% Jun 27	7	optional, final latency
50 stations instead of 25 (with gradients)	-15.0% mean, 5 of 5 better	5	descriptive test; not adopted

Mean URE ratio to NGA on the August days: 0.94 at the baseline (tied with NGA on Aug 2), 0.88 after Earth radiation and thrust, 0.75 after sub-daily EOP and 0.73 after elevation weighting. The 3D ratio improved less, because the along-track component improved least.

Evidence: the recorded evaluations `Opus_Fork/results/arm_*.json`; figure data `research/report_rev4/figures.py`; `Opus_Fork/experiments/networks/RESULTS.md`.

10. What did not work

Tried	Outcome	Decision
Free ECOM-7 (twice-per-rev D terms)	URE +29.5% (Aug 2), +16.9% (Aug 3)	Rejected
ECOM-7 with 2 nm/s2 priors	-2.9 to -6.1% on Aug 2-5, +4.5% on Aug 7	Rejected (failed hold-out)
15-degree cutoff instead of 10	Held-out mean +0.06%	Rejected
Per-station empirical reweighting (W1/W2)	Gains only on the day they were tuned on	Not adopted
Estimated D0 on 6 h arcs	Better Aug 2, worse Aug 3	Not adopted
GPT3 empirical gradients	5 of 5 dates better, but mean -0.61% (bar: 1%)	Inconclusive
Hourly wet delay (24 h)	-1.4% on one studied day (diagnostic)	Not adopted
Desai-Sibois sub-daily EOP	Within 3% of ORTHO_EOP in a census	Not used
Box-wing a-priori solar model	Census only	Unvalidated

Diagnostics with small predicted effects

- **Removing satellite antenna variations** worsened URE by 25%: they matter.
- **Specified atmospheric perturbations** had small predicted effects in a local linear transfer model, against a 3.165 cm baseline (SD-3): independent 10 hPa station pressure 0.035 cm (1.1%); common 10 hPa 0.068 cm (2.2%); S2 climatology 0.040 cm (1.3%); 1 mm mapping error at 10 degrees 0.0007 cm. Station-tide figures refer to the 12-hour scale amplitude, not URE. The method was calibrated on one real refit; these do not rule out all such errors.
- **Station gradients:** assuming independent errors of 0.5 mm per north and east component, the same model predicted 0.73 cm URE. That was an assumption, not a measured field, and it led to the gradient tests (page 13).

Why rejected items stay rejected

Each test had a written rule before it ran. A rejected configuration is not retuned on the same days. For example, a "use ECOM-7 only when the swing is present" rule was proposed and refused.

Evidence: `docs/consolidation/configurations.md` sections 3-5; `Opus_Fork/results/*_RESULTS.md`; `Opus_Fork/experiments/atmosphere/RESULTS.md`; `Opus_Fork/results/CORRECTIONS.md`.

11. Untouched-date validation

Why untouched dates (30 September). Every August day had shaped the model choices. So before any download, a protocol froze the pipeline, the success rule and a seeded random stream of dates from 1 June to 27 August. Dates were taken from the stream in order and replaced only for data reasons, such as incomplete navigation files.

Date	Our URE (cm)	NGA URE (cm)	URE ratio	Our 3D (cm)	NGA 3D (cm)
Jun 6	2.911	3.804	0.765	9.17	10.30
Jun 20	3.395	3.911	0.868	9.45	10.22
Jun 23	2.964	4.306	0.688	8.42	10.63
Jul 6	2.914	3.798	0.767	9.06	10.55
Jul 23	3.127	4.210	0.743	8.77	10.82
Mean	**3.062**	**4.006**	**0.766**	**8.97**	**10.50**

Result: below NGA in URE and in 3D on all five dates.

- Radial and cross-track are better on every date.
- Along-track is worse than NGA's on Jun 6, Jul 6 and Jul 23.

Qualifications:

- **The repair.** Jun 23 first stopped on a code defect: a numerical derivative crossed the edge of an antenna calibration table. It was fixed and refitted under a documented amendment. The original as-registered result is therefore "not established"; this is the transparently repaired analysis.
- **Model selection.** The configuration was chosen on August days using IGS-scored outcomes. These dates test transfer, but they are one season and one network.
- **Exposure.** These dates were untouched for this test only. The later gradient and station tests reused them, so those are new-arm contrasts on exposed dates (page 22).
- **Independent reproduction.** A separate scorer with its own frame and Earth-orientation code (`validation/oracle`) reproduces all five URE and 3D scores within 0.1 mm (mean ratio 0.766352). Astra's review did the same.

Evidence: `Opus_Fork/experiments/untouched_dates/` (`PROTOCOL.md`, `RESULTS.md`, `scores.json`); `research/astra_followup_review_20260930/`; `validation/oracle/`.

12. Tropospheric gradients

Why gradients. The linear transfer model predicted about 0.7 cm URE from station gradients of 0.5 mm, an assumed error size (page 11). We therefore tested external, fixed gradient fields:

- **GRAD operational:** TU Wien weather-model gradients, 6-hourly [R6, R12];
- **GPT3:** an empirical climatology [R5].

Test	GRAD operational	GPT3
Aug 3 (studied day; selection)	-4.16% URE	-1.50%
Five dates outside the selection (pre-registered)	-1.15, -1.72, -2.14, -0.42, -4.07%: mean -1.90%, 3D -2.39%	mean -0.61%
Reading	**validated on these dates**	inconclusive
Jun 27, a fresh date (matched default arm added afterwards)	-3.83% (2.859 to 2.750 cm; NGA 3.795)	-

Size and spread. The mean gain is 0.59 mm URE; the smallest daily gain is 0.12 mm (Jul 6). 109 of 154 satellite-days improved in 3D; the worst worsened by 1.98 cm (G24, Jun 6).

Controls:

- the first-iteration trajectories are bitwise equal with and without gradients, since gradients enter only the observations;
- the independent scorer reproduces URE and 3D within 7 micrometres;
- a linear prediction made before any Aug 3 fit matched the refits.

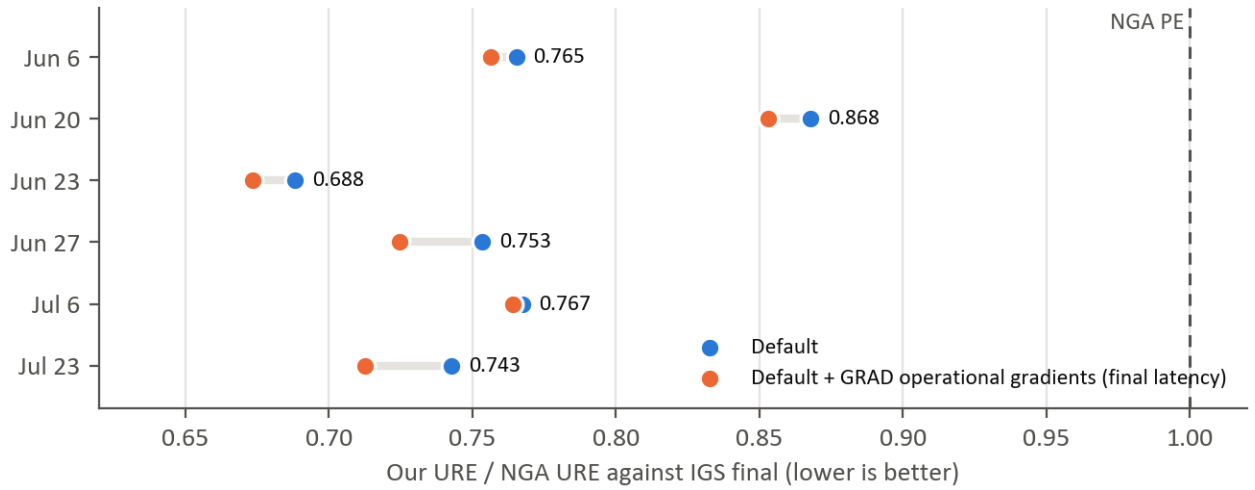
Adoption (user, 3 Oct): optional, labelled, final latency only.

- **Why not rapid.** The last GRAD file an arc needs carried a server Last-Modified time about 40 h after the arc ended. That was read later; it is not an observed first availability or a verified issue time.
- **What the default does.** It stays without gradients, for anything that claims rapid latency.
- **Labelling.** Products made with gradients name the model in the header and use a local "FIN" file label, which certifies no IGS class or delivery time.
- **The rapid route** would be TU Wien's forecast gradients, which need credentials.

About Jun 27. A "sequentially amended fresh-date comparison": the gradient arm was scored first, and the matched default arm was added afterwards.

Evidence: `Opus_Fork/experiments/gradients/` (`VALIDATION_PROTOCOL.md`, `VALIDATION_RESULTS.md`, `NEW_DATE_RESULTS.md`).

13. Fresh dates at a glance



URE ratio to NGA PE on every fresh date, for the default (blue) and the default with GRAD operational gradients (orange). Every point is left of the dashed line: closer to IGS final than NGA PE.

Date	Default ratio	+ gradients ratio	Gradient change
Jun 6	0.765	0.757	-1.15%
Jun 20	0.868	0.853	-1.72%
Jun 23	0.688	0.674	-2.14%
Jun 27	0.753	0.725	-3.83%
Jul 6	0.767	0.764	-0.42%
Jul 23	0.743	0.713	-4.07%

What the gradients change. They improve the radial component on every date and the 3D error by 1.6-5.0%. They do not change the overall picture: we are 13-33% below NGA in URE on every fresh date, with or without them.

What stays the same. The along-track component is where NGA remains better on several dates, and gradients help it only a little (-0.8% on Jun 27). The largest remaining gap to NGA is therefore along-track. Force modelling (eclipse attitude, empirical terms) and observability are the next candidates to test; the atmosphere is not ruled out.

14. How often do we beat NGA?

The URE ratio says how much closer we are on average. The **time share** says how often. It counts the 15-minute epochs at which our constellation URE is below NGA's, and separately the individual satellite-epochs. All values are computed with the independent scorer on each comparison's own satellite-epochs. Each one reproduces the production URE within 0.1 mm.

Studied days Aug 2-5, 7 (pooled)	Mean URE ratio to NGA	15-min epochs below NGA	Satellite-epochs below NGA
Baseline	0.942	304/480 (63.3%)	50.8%
+ Earth radiation and thrust	0.876	367/480 (76.5%)	54.7%
+ sub-daily EOP	0.753	442/480 (92.1%)	63.5%
+ elevation weighting (default)	0.727	456/480 (95.0%)	65.0%

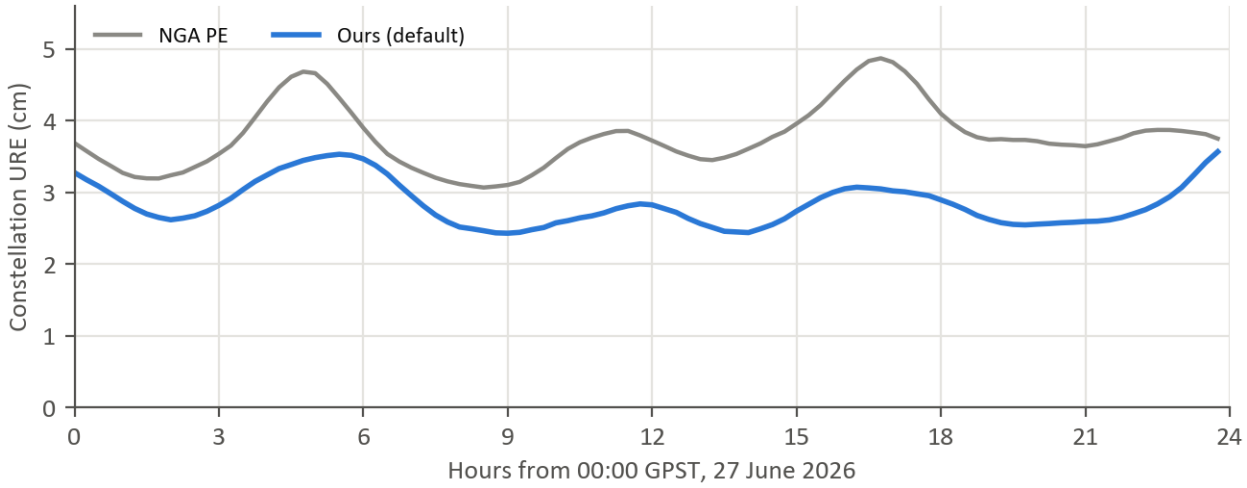
Rejected configurations	Mean URE ratio to NGA	15-min epochs below NGA	Satellite-epochs below NGA
Free ECOM-7 (Aug 2, 3)	1.138	59/192 (30.7%)	50.1%
ECOM-7 with priors (Aug 2-5, 7)	0.854	395/480 (82.3%)	56.1%
15-degree cutoff (Aug 2-5, 7)	0.745	451/480 (94.0%)	64.4%
Per-station reweighting W2 (Aug 2)	0.982	64/96 (66.7%)	51.8%

What the shares show:

- **The baseline was barely ahead.** On the studied days it was below NGA at 63% of epochs; the default is below NGA at 95%. The sub-daily EOP made the largest single difference in how often we win, from 76% to 92%.
- **Single satellites are close to a coin flip.** The satellite-epoch share stays near 51-65% because, for one satellite at one moment, the two products trade places often. The constellation advantage comes from consistently smaller errors spread over all satellites, not from winning every comparison.
- **The two shares can order differently.** Free ECOM-7 was below NGA at only 31% of epochs but at 50% of satellite-epochs.
- **The 15-degree cutoff was not bad against NGA.** It was rejected because it did not beat the default on held-out days (mean +0.06%).

Scope. These are sampled 15-minute epochs on studied summer days, not continuous time and not a probability for other dates. IGS final is the reference.

15. Hour by hour: one fresh date



Constellation URE every 15 minutes on 27 June 2026, ours (default configuration, blue) and NGA PE (gray), both against IGS final. Revision 5 corrects the hour axis by 10 s (the arc starts at 23:59:50); the values are unchanged.

Fresh date	Default: ratio	Default: epochs below NGA	+ gradients: epochs below NGA
Jun 6	0.765	89/96 (92.7%)	90/96 (93.8%)
Jun 20	0.868	84/96 (87.5%)	85/96 (88.5%)
Jun 23	0.688	88/96 (91.7%)	88/96 (91.7%)
Jun 27	0.753	96/96 (100%)	96/96 (100%)
Jul 6	0.767	83/96 (86.5%)	83/96 (86.5%)
Jul 23	0.743	90/96 (93.8%)	92/96 (95.8%)
All six fresh dates		**530/576 (92.0%)**	**534/576 (92.7%)**

On the fresh dates our constellation URE is below NGA's at 92% of the sampled epochs; 86.5% on the worst date and every epoch on Jun 27. Individual satellite-epochs are below NGA 63.0% of the time (63.8% with gradients). Astra's independent check reproduces these counts.

What the figure shows:

- On Jun 27 both products have twice-daily bumps, ours smaller.
- **The arc edge.** Our error rises in the last hour, where the batch fit has data on one side only. Overlapping or longer arcs, which are not implemented, would address that.

Scope. These are sampled constellation epochs on six summer dates, not continuous time, not a per-satellite probability and not a forecast. The epochs are not independent trials.

16. Inputs and rapid latency

The objective asks for inputs that could plausibly be available within about a day. Every run so far used files downloaded afterwards. A published model existing in advance is different from the exact file versions used here having existed at a cut-off; none of those versions was verified for a cut-off.

Input	Used for	Rapid-suitable?
Station observations, daily 30 s files	Fit	Plausible: daily files normally appear within hours; receipt times not recorded
Broadcast navigation, day and previous day	Fit (starting orbit, satellite selection)	Plausible: the message is real time; the merged daily file's receipt time not recorded
IERS daily EOP (`finals2000A.all`)	Fit, scoring, export	**Not as used.** A later file with final values; a rapid product would use predicted values. Effect not measured
Published models: IERS sub-daily EOP, DE440s, EGM2008, FES2014b	Fit	Yes: analytical models published in advance
Dated files: IGS20 coordinates and ANTEX, IGS satellite metadata	Fit	**Retrospective, unverified.** The coordinate product is cumulative and updated; the metadata file is from September
TU Wien GRAD operational gradients	Optional fit model	**No.** Last-Modified about 40 h after the arc; final latency
TU Wien GPT3 gradients	Not adopted	Yes (a climatology)
IGS final, NGA PE	Scoring only, never read by the fit	Not applicable

What has not been shown

- **No as-of replay.** We have not established which exact bytes of each input existed by a given cut-off. That replay is the next step for the rapid objective.
- **Predicted EOP and other vintages.** Their effect on the orbit has not been measured.
- **Delivery time.** Guarded runs that pause for machine load and resume are not a delivered product within a fixed deadline. Active computing and elapsed delivery time would have to be reported separately.

17. Results using non-rapid data

These results are kept apart from the default because something in them would not be available, or would not be legitimate, in a rapid station-only product.

Result	What is not rapid or not independent	Where it stands
Default + GRAD operational gradients (pages 13-14)	GRAD files about 40 h after the arc	Optional final-latency model; default unchanged
Station tests S17 and S50 (pages 19-20)	Use GRAD gradients	Descriptive final-latency contrasts; nothing adopted
Block-mean D0 calibrated from IGS rapid (6 h arcs)	Calibrated from a reference orbit	Superseded; never in the default
ECOM-7 with priors	Prior width from fits to IGS final	Rejected
Reference-fitted force tests (ECOM5 to 0.8 cm)	Fitted to IGS positions	Diagnostics only, never fit inputs
Censuses of IGS orbits (swing, box-wing, sub-daily EOP)	Fitted to IGS final	Diagnostics only
Weighting exponent of elw13	Own residuals (independent) but adopted on IGS-scored outcomes	In the default; disclosed as "reference-scored selection"
Every result in this report	Retrospective EOP, coordinates and metadata	See page 17

Rules we follow

- **Fit inputs and model selection are recorded separately.** A model can have independent inputs and still have been chosen by IGS-scored results. That is a selection effect, and we say so.
- **Reference-derived priors are "reference-informed".** They disqualify a reference-independence claim.
- **Reference products are only ever read by scoring code.** The fit code refuses to open them.

Evidence: `Opus_Fork/results/CORRECTIONS.md` (C02); `docs/consolidation/configurations.md` (legend and section 5); `docs/DATA_SOURCES.md`.

18. Station tests: design

The user asked for two tests, both with the best configuration (default + GRAD operational gradients, so final latency), on the five fresh dates that already had matched 25-station fits (V-op). The protocol was pinned before any station was chosen (`Opus_Fork/experiments/networks/PROTOCOL.md`, amendments A1-A4).

S17: 17 selected IGS stations near NGA's monitor sites. For each publicly described NGA or USAF site, the nearest usable IGS station: MOBS, UNSA, BRUX, BSHM, GLPS, NRC1, WHIT, PIMO, THTG, HRAO, SYDN, NKLG, NLIB, NIST, SGOC, KOKB, MAJU. This is a proxy for the public site list, not NGA's data or estimator.

- Eight are more than 1,000 km from their site: UNSA 1,304, BSHM 1,664, GLPS 1,317, PIMO 2,576, SYDN 2,245, NKLG 2,819, NLIB 1,772 and SGOC 1,779 km.
- The estimator needs BRUX as its clock datum, so (amendment A1, before any fit) BRUX (404 km) replaced the nearest station HERS (141 km) for Hermitage.

S50: the 25-station network doubled. The 25 plus 25 stations added one at a time, each the station farthest from all those already chosen: STJ3, JOG2, TOW2, BSHM, LAUT, KIRU, UNSA, ALGO, IQAL, DARW, SEYG, SCOR, ALBH, POHN, MAL2, DUBO, CEBR, ALIC, QAQ1, JOZE, CKIS, KARR, WHIT, SYDN, NIST. Jun 6 had 49 (no DAV1) and Jul 6 had 49 (no GUAM); the matched 25-station arms had 24 on those dates.

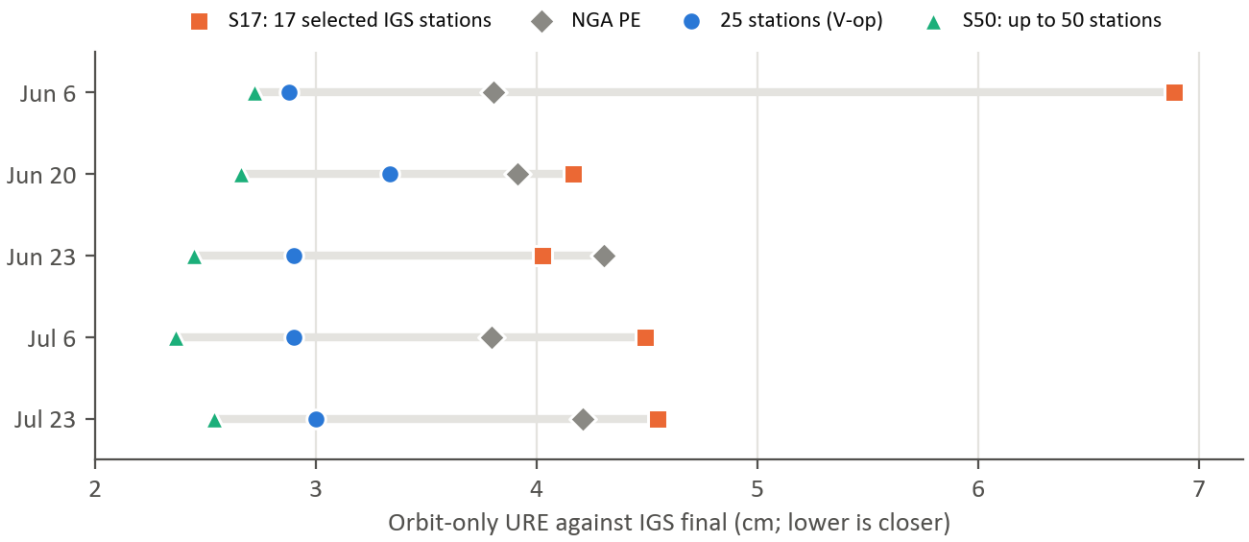
What limits station choice (293 stations listed at BKG on all five dates):

Requirement	Reason
IGS20 coordinate solution valid on the date	123 sites have none; 13 more had expired solutions
No post-seismic deformation model	20 sites excluded (e.g. LPGS, FAIR, DAEJ, WGTN, DGAR)
P(Y)-code signal pair (C1W/C2W)	10 receivers without one (e.g. ASCG) would need code-bias corrections
Supported phase-shift header; exact antenna calibration	9 and 1 rejected
GRAD gradient coverage (optional model only)	4 sites

These are limits of the current preparation code, not of the stations.

A leaner solve for 50 stations. The first S50 fit stopped at its 512 MiB workspace check. Instead of raising the limit, an exact option folds each phase ambiguity out of the square-root system once its arc has ended (amendment A2). A real-data parity fit agreed with the standard solver to 0.01 mm (limit 1 mm).

19. Station tests: results



Orbit-only URE against IGS final on each date, for S17 (17 stations), NGA PE, our 25 stations and S50 (up to 50 stations). All our arms use the default plus GRAD operational gradients.

Date	NGA PE (cm)	S17 (cm)	25 stations (cm)	S50 (cm)	S50 vs 25
Jun 6	3.80	6.89	2.88	2.72	-5.4%
Jun 20	3.91	4.17	3.34	2.66	-20.2%
Jun 23	4.31	4.03	2.90	2.45	-15.6%
Jul 6	3.80	4.49	2.90	2.36	-18.5%
Jul 23	4.21	4.55	3.00	2.54	-15.3%
Mean	**4.01**	**4.83**	**3.00**	**2.55**	**-15.0%**

- **S17 does not stay closer to IGS final than NGA.** Below NGA on 1 of 5 dates; mean ratio 1.22; below NGA at 40% of 15-minute epochs (25 stations: 91%). URE is 62% higher than with 25 stations.
- **S50 "improves"** under the pre-registered rule: better on 5 of 5 dates, mean -15.0%, 3D -18.4%, radial lower on every date. Below NGA at 99.6% of epochs. The worst satellite got 4.2 cm worse in 3D (G26, Jun 20).
- **What it does not show.** Both tests change geometry and composition as well as count, so they do not say which matters, or why NGA does better with a network of similar size.

Checks. All five controls passed: independent rescoring of every arm and of NGA, bitwise-equal first trajectories, provenance and model identity, and the exact scored population. One control first failed by 0.015 mm because the two codes used different, equally valid ERFA pole-position methods; the method was matched and the 0.1 mm limit kept (page 21).

Evidence: `Opus_Fork/experiments/networks/RESULTS.md`; `networks_analysis.json`; `Opus_Fork/results/networks_*.json`.

20. Engineering and review

Running on one PC

- **Guards.** Campaign fits, evaluations and report builds run inside resource guards (Windows Job Objects) that cap memory, CPU, threads and time. The fork8 profile holds its GPU telemetry for 10 s (a documented limitation). Small support analyses may run unguarded within the same machine limits.
- **The campaign lease.** Lease-holding controllers and guards take it, so two such campaigns cannot run at once; other controllers are not wired to it.
- **Resumable chains.** Fits resume after shutdowns or machine-load stops; completed workers are reused after hash checks.

Checking our own work

- **Pinned protocols.** Every adoption test had a protocol pinned in its campaign records before it ran. Amendments are appended and say whether outcomes existed.
- **Independent scorer.** `validation/oracle` has its own frame, EOP and sub-daily code; it shares ERFA, inputs and conventions. It now offers both ERFA pole-position methods: the series method (default, a cross-method check; 0.2 mm at most) and the angles method used by production, for strict reproduction within 0.1 mm.
- **Tests.** 427 unit and regression tests, including negative controls: 425 run directly and 2 check the guard from inside it, all passing on 8 Oct. Some real-data tests skip when local data are absent.
- **The correction index.** `Opus_Fork/results/CORRECTIONS.md` lists every correction to an earlier claim.

Independent reviews

Astra (OpenAI Codex) reviewed each checkpoint, reproduced the untouched-date ratios, the gradient gains and the 92% share, wrote a principal-investigator review before publication, and reviewed the station-test controls before any score was read. A separate reviewer (Sol) found the antenna-table derivative defect. Reviews are in `research/`.

Defects found and fixed (examples)

- **SP3 export.** A sub-daily-EOP solution was exported with the wrong rotation (about 5 cm); export could also overwrite another product. Both fixed, with tests.
- **Derivative stencil.** A numerical derivative crossed an antenna-calibration boundary (one link of 2,777).
- **Integrity.** Truncated arrays could pass a hash check; a repeated model install could apply weighting twice; the scorer's checker could exit 0 after a failure. All repaired with regression tests.

21. What these results do and do not show

The contribution. A reproducible, resource-bounded, self-written estimator with a carefully recorded development and validation process, and a bounded retrospective performance demonstration. Square-root least squares, ECOM and network processing are established methods; no new estimation method is claimed.

Input role	What it is
Direct observations	Station code and phase (RINEX 3)
Initialization	Broadcast navigation
Fixed external models and calibrations	IGS20 coordinates, igs20.atx, IGS satellite metadata, IERS EOP, FES2014b, DE440s, EGM2008
Fitted constraints	Loose state priors, constrained wet delay, BRUX clock datum
Reference products	IGS final and NGA PE: scoring and model selection only, never fit inputs

Relative, not absolute. "About 3 cm" is orbit-only URE against IGS final; the 3D RMS is about 9 cm. Shared geodetic inputs can correlate our errors with the reference. Clocks are not estimated as a product, so this is not a positioning or full signal-in-space result.

Dates	First use	Later exposure
Aug 2-5, 7	Development and model selection	All later work: new-arm contrasts
Jun 6, 20, 23, Jul 6, 23	Untouched validation of the default (Jun 23 repaired)	Gradients (3 Oct), station tests (8 Oct)
Jun 27	Fresh-date gradient run (4 Oct)	Matched default added after the gradient result

Small effects and spread. Five dates do not give a general success probability, and epochs or satellites are not independent replicates. Aggregate gains coexist with local losses: the gradient model worsened one satellite by 1.98 cm, S50 one by 4.2 cm. SP3 files round positions to 1 mm.

22. Limitations and next steps

Limitations

- **Retrospective inputs.** No rapid-deadline replay; predicted-EOP effect not measured.
- **Along-track.** NGA PE is still better on several dates. The 12-hour breathing (0.16-0.37 ppb) is unexplained.
- **Physical models.** Nominal yaw through eclipses; fixed radiation and thrust parameters and masses; diagonal weighting; libration omitted. Improvements narrow hypotheses; they do not identify unique causes.
- **Numerical assurance.** Reintegration checks cover synthetic and short cases, not the full adopted model at real boundaries. The Earth-radiation variational derivative omits the radial force-magnitude gradient (a recorded approximation, not yet bounded). The covariance is conditional on fixed coordinates and supplied sigmas.
- **Release integrity.** Evaluation and export check solution hashes and fixture bindings, but a strict check of the full model, fixture and EOP identity is not yet enforced for new products (PI-08). The fork guard can release its admission before killed child processes exit (B16).
- **Reproduction.** No isolated end-to-end rehearsal of the public release yet; tests needing local data skip.
- **Breadth.** GPS only; float ambiguities; one season; fixed networks; Windows-only guards.

Recommended next steps, in order

1. **Freeze the claim and the configuration** of any paper or release, with this page as its scope.
2. **Close the integrity items** PI-08 and B16, each with negative controls.
3. **Rehearse the public release in isolation:** one fit from frozen raw inputs to an exported SP3, outside this PC.
4. **A frozen untouched-date test of a 50-station default** without gradients, if a denser network is wanted.
5. **Rapid as-of replay** with predicted EOP and dated inputs; forecast gradients if credentials allow.
6. **Along-track:** eclipse attitude and empirical terms, each under its own frozen protocol.

Why this order. The retrospective bar is met; what limits the claims now is integrity, reproducibility and latency, more than another model term.

23. Evidence index

Topic	Main evidence
Setup, data sources, specification, public release	SETUP.md; docs/DATA_SOURCES.md; docs/REPRODUCING_WITH_AN_LLM.md; docs/KNOWN_LIMITATIONS.md
Every configuration and its status	docs/consolidation/configurations.md
Corrections to earlier claims	Opus_Fork/results/CORRECTIONS.md
First phase (6-13 Sep)	STAGE0_RESULTS.md to STAGE5B_RESULTS.md; YELL_GUAM_RESULTS.md; ARC_LENGTH_RESULTS.md; D0_POD_RESULTS.md; D0_DAY215_RESULTS.md; GUAM_GAP_RESULTS.md
24 h arcs and 25 stations	claude/roadmap/STEP3_RESULTS.md; claude/roadmap/BENCH_RESULTS.md; claude/roadmap/STATIONS_RESULTS.md
NGA bar and baseline	Opus_Fork/results/BASELINE.md
Forces and EOP	Opus_Fork/results/T2_1_ERP_RESULTS.md; IIIA_AUG345_RESULTS.md; SDEOP_RESULTS.md; ECOM7_RESULTS.md; ECOM7P_RESULTS.md; ECOM7P_HOLDOUT_RESULTS.md
Weighting and sensitivity	Opus_Fork/results/ELEVATION_RESULTS.md; SENSITIVITY_RESULTS.md; WEIGHTS_W1.md; WEIGHTS_W2_AND_CAUSE.md
Untouched dates	Opus_Fork/experiments/untouched_dates/ (PROTOCOL.md, RESULTS.md, scores.json)
Atmosphere and gradients	Opus_Fork/experiments/atmosphere/RESULTS.md; Opus_Fork/experiments/gradients/
Time shares and figures	Opus_Fork/experiments/gradients/ure_time_share.json; research/report_rev4/; research/report_rev5/
Station tests	Opus_Fork/experiments/networks/ (PROTOCOL.md with A1-A4, RESULTS.md, networks_analysis.json, scoring_index.json)
Independent reviews	research/*review*/REVIEW.md, notably research/pi_review_20261006/ and research/astra_network_review_20261008/

Scores quoted in this report come from the recorded evaluations in `Opus_Fork/results/` (`arm_*.json`, `gradval_*.json`, `newdate_*.json`, `networks_*.json`). Paths are relative to the repository root.

24. Sources, terms and revision

References

- **R1.** IGS, Products: rapid and final latency. igs.org/products. Rechecked 6 Oct 2026 (Astra documentation review) for the 17-hour rapid description.
- **R2.** GROOPS cookbook, GNSS network processing; architecture context only. Cookbook. Rechecked 6 Oct 2026.
- **R3.** Petit G, Luzum B (eds) (2010) IERS Conventions (2010). IERS Technical Note 36, Verlag des BKG.
- **R4.** Arnold D et al. (2015) CODE's new solar radiation pressure model for GNSS orbit determination. J Geod 89:775-791. doi:10.1007/s00190-015-0814-4.
- **R5.** Landskron D, Bohm J (2018) VMF3/GPT3: refined discrete and empirical troposphere mapping functions. J Geod 92:349-360. doi:10.1007/s00190-017-1066-2.
- **R6.** Landskron D, Bohm J (2018) Refined discrete and empirical horizontal gradients in VLBI analysis. J Geod 92:1387-1399. doi:10.1007/s00190-018-1127-1.
- **R7.** Steigenberger P, Thoelert S, Montenbruck O (2020) GPS III Vespucci: results of half a year in orbit. Adv Space Res 66:2773-2785. doi:10.1016/j.asr.2020.03.026.
- **R8.** Park RS et al. (2021) The JPL planetary and lunar ephemerides DE440 and DE441. Astron J 161:105. doi:10.3847/1538-3881/abd414.
- **R9.** Pavlis NK et al. (2012) The development and evaluation of the Earth Gravitational Model 2008 (EGM2008). J Geophys Res 117:B04406. doi:10.1029/2011JB008916.
- **R10.** Lyard FH et al. (2021) FES2014 global ocean tide atlas. Ocean Sci 17:615-649. doi:10.5194/os-17-615-2021.
- **R11.** ERFA, derived from IAU SOFA: `xy06` (series) and `xys06a` (angles) CIP methods. liberfa/erfa.
- **R12.** TU Wien VMF data server, GRAD gradients. vmf.geo.tuwien.ac.at.

Full data sources and citation requests: `docs/DATA_SOURCES.md`.

Terms

- **URE:** the orbit-only user range error defined on page 2. **RAC:** radial, along-track, cross-track.
- **EOP:** Earth orientation parameters. **ECOM:** the Empirical CODE Orbit Model. **elw13:** phase weighting 9 mm/sin(e)^1.3. **NGA PE:** the National Geospatial-Intelligence Agency's precise ephemeris.
- **Untouched date:** a date drawn by a pre-registered rule, whose data informed no earlier choice.

This revision

- **What it covers.** Revision 5 adds the station tests, the leaner 50-station solve, the SP3 deliverables, the corrections from Astra's 6 Oct documentation and principal-investigator reviews, and the scope page.
- **How it is kept.** Revision 4 is archived byte-for-byte in `research/report_history/revision4/`. The manifest records the source, evidence and output hashes, and the visual QA of every page.
- **What this update did not do.** No experiment was rerun for this document. New figures come from recorded evaluations through `research/report_rev5/`.